

PHYSICS – BOARD QUESTION PAPER (JULY 2024)**Time: 3 Hours****Max. Marks: 70****General Instructions:**

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Calculator is not allowed.
6. Figures to the right indicate full marks.
7. For MCQs, only the first attempt will be considered.

Physical Constants:

$$g = 9.8 \text{ m/s}^2$$

$$\pi = 3.142$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

SECTION – A**Q1. Select and write the correct alternatives: (10M)**

i. Atoms having same number of protons but different neutrons are called:

(A) Isotopes (B) Isobars (C) Isotones (D) Isomers

ii. The molecules on the surface of a liquid have:

(A) Minimum kinetic energy (B) Minimum potential energy

(C) Maximum kinetic energy (D) Maximum potential energy

iii. The unit henry is equal to:

(A) watt (B) ohm-second (C) dyne (D) Wb/m²

iv. LED emits light when:

(A) Reverse biased (B) Depletion region widens

(C) Holes and electrons recombine (D) Junction becomes hot



v. Cyclotron is used to accelerate:

- (A) Neutral particles (B) Negatively charged particles
(C) Positively charged particles (D) All types of particles

vi. An electron, proton, α -particle and hydrogen atom have same kinetic energy. The longest de-Broglie wavelength is for:

- (A) Electron (B) Proton (C) α -particle (D) Hydrogen atom

vii. For a perfectly black body with coefficient of emission e and transmission t :

- (A) $e=0, t=1$ (B) $e=1, t=0$ (C) $e=\frac{1}{2}, t=\frac{1}{2}$ (D) $e=|t|$

viii. Ratio of emissive power at 927°C and 327°C is:

- (A) 2:1 (B) 4:1 (C) 8:1 (D) 16:1

ix. In a series LCR circuit, if phase difference between voltage and current is 45° , power factor is:

- (A) 0.6071 (B) 0.7071 (C) 0.5000 (D) 1.0000

x. When an air column in a pipe closed at one end vibrates such that two nodes are formed, frequency is:

- (A) Two times fundamental (B) Three times fundamental
(C) Four times fundamental (D) Five times fundamental

Q2. Answer the following: (8M)

i. Radius of gyration of a solid sphere of radius R about its diameter.

ii. Write differential equation of linear SHM.

iii. State any one method of polarization of light.

iv. What is resistance of an ideal voltmeter?

v. What are eddy currents?

vi. What do you mean by logic gate?

vii. A body of mass 0.2 kg performs SHM. Restoring force = 0.4 N at displacement 8 cm.
Find force constant.

viii. Work done in bringing $5\text{ }\mu\text{C}$ from infinity to point at 400 kV.

SECTION – B

Attempt any Eight questions. (16M)



- Q3. Show that average energy per molecule is directly proportional to absolute temperature.
- Q4. Explain cyclic process with neat and labelled p–V diagram.
- Q5. Distinguish between progressive waves and stationary waves.
- Q6. Explain Biot–Savart law with suitable diagram.
- Q7. Draw neat and labelled diagram of Van de Graaff generator.
- Q8. State formula for (i) Average power in LCR circuit (ii) Q-factor.
- Q9. Explain working of a transformer with neat labelled diagram.
- Q10. A galvanometer of resistance 100Ω and full-scale current 0.2 mA . Find resistance to convert it into a $0\text{--}10\text{ V}$ voltmeter.
- Q11. An electron in hydrogen atom stays in second orbit for 10^8 s . Velocity = $1.07\times 10^6\text{ m/s}$, radius = $2.14\times 10^{-10}\text{ m}$. Find number of revolutions.
- Q12. A torque of 400 Nm produces angular acceleration 20 rad/s^2 . Calculate moment of inertia.
- Q13. A bar magnet of moment of inertia 500 g cm^2 oscillates with period 3.142 s in horizontal field $4\times 10^{-5}\text{ T}$. Find its magnetic moment.
- Q14. A telescope has objective diameter 2.5 m . Find angular resolution for $\lambda = 6000\text{ \AA}$.

SECTION – C

Attempt any Eight questions. (24)

- Q15. Define surface tension. Obtain relation between surface tension and surface energy.
- Q16. Show that all harmonics are present in case of a stretched string.
- Q17. Derive expression for impedance of a series LCR circuit.
- Q18. What is Curie temperature? Distinguish between diamagnetic and paramagnetic substances.
- Q19. Obtain balancing condition for Wheatstone's bridge.
- Q20. What is ionization energy? Using formula for energy of electron, derive expression for wavelength of spectral lines in hydrogen.
- Q21. What is voltage regulation? Explain working of Zener diode as voltage regulator.
- Q22. A spherical oil drop falls at 9.8 cm/s . Density of oil = 0.9013 g/cm^3 , density of air = 0.0013 g/cm^3 , viscosity of air = $1.8\times 10^{-4}\text{ poise}$. Calculate radius.
- Q23. A search coil with 2000 turns and area 1.5 cm^2 is moved out of a 0.6 T field in 0.2 s . Calculate induced emf.
- Q24. For SHM amplitude 10 cm , find distance from mean position where kinetic energy is three times potential energy.
- Q25. When 2×10^{10} electrons shift from one conductor to another, potential difference becomes 20 V . Find capacitance.



Q26. Magnetic field at centre of circular loop (radius 12 cm) is 6×10^{-4} T. Find magnetic moment.

SECTION – D

Attempt any Three questions. (12M)

Q27. Derive expression for minimum speed required to perform stunts in well of death.

Q28. Explain concept of positive and negative work with diagrams.

Q29. Deduce expression for molar specific heat of monoatomic gas at constant volume.

Q30. What is photoelectric effect? Describe experiment with neat circuit diagram.

Q31. Derive formula for Brewster's angle. Green light of wavelength $5100 \text{ \AA}$ forms 10 fringes in 2 cm on screen 2 m away. Find slit separation.

